

Abhinandan

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EXPERIENCE

Founding Software Engineer

Sept 2025 – Present

Browser

Remote, San Francisco

- Built production agent/runtime infrastructure for browser automation: Chrome MV3 recorder, CDP execution, AX/DOM element capture, cross-iframe support, streaming ReAct execution, multi-tab orchestration, and safe parallel tool calls.
- Cut cost per automation run $\sim 3\times$ across **700K+ monitored production LLM calls** using compact traces, context compression, prompt caching, and dynamic Haiku/Sonnet/GPT-5 routing.
- Shipped a zero-LLM replay engine with variable-driven tool-call templates and stateful AI fallback that resumes failed runs mid-execution instead of restarting them.
- Built a proactive DevRel/content system that turns product workflows into documentation portals, quickstarts, long-form guides/books, and demo assets, with self-healing regeneration when the underlying UI changes.

Founding Software Engineer

Jan 2025 – Jul 2025

Cynos Nexus

Noida, India

- Built FastAPI retrieval and ingestion services with vector search, Redis/BullMQ workers, structured document processing, and production API serving.
- Automated containerized deployment and CI/CD for AI services using Docker, GitHub Actions, Nginx, and cloud infrastructure.

Contract Software Engineer

Sept 2024 – Dec 2024

Etkin.ai

Remote, Turkiye

- Built a LangChain + pgvector retrieval system; improved recall@5 from **0.71 to 0.82** on a 300-query labeled evaluation set.
- Built Celery-backed ingestion pipelines and production PDF/report generation systems; reduced bulk processing time by **40%**.

Open Source Contributor

June 2024 – Aug 2024

HeroUI

Y Combinator S24 startup

- Received a **personal offer from the CEO** to join HeroUI (prev. NextUI) after making **open-source contributions**.

SWE Intern

Jan 2024 – Apr 2024

SkilledUp

Noida

- Engineered a context-aware AI code assistant for the core learning dashboard using FastAPI and OpenAI.

SELECTED PROJECTS

Mini LLM Inference Runtime | *Python, scheduling, KV-cache management*

- Implemented request admission, continuous batching, token-budget scheduling, chunked prefill, decode iteration, and fixed-block KV-cache allocation to model the core control plane of modern LLM serving engines.
- Built workload simulations comparing static vs. continuous scheduling under heterogeneous prompt/decode lengths, tracking request latency, token utilization, and decode starvation behavior.

AgentFlow-Pro | *PyTorch, TRL, DAPO, PRM, LoRA* | [GitHub](#)

- Rebuilt the ICLR 2026 AgentFlow architecture as a local Qwen3-8B Planner $\rightarrow$ Executor $\rightarrow$ Verifier $\rightarrow$ Memory system with constrained planning and sandboxed Python/SymPy tools.
- Trained the planner with DAPO + a learned Qwen3-0.6B Process Reward Model and measured **+5.0 pts on GPQA-Diamond** (40.0% $\rightarrow$ 45.0%) with a leakage-free evaluation pipeline.

smartmemo | *PyTorch, sentence-transformers, FAISS, SQLite* | github.com/awesome-pro/smartmemo

- Built a semantic memory/cache for LLM agents where embedding retrieval proposes candidates and a learned pair-equivalence classifier decides whether memory reuse is safe.
- Trained classifier-v2 on 16,576 pairs across 9 domains, improving precision by +30 points at equal recall versus cosine similarity, with feedback export, bad-hit detection, gated retraining, and WAL-backed SQLite hardening.

TECHNICAL SKILLS

ML / Inference: PyTorch, Hugging Face, vLLM, SGLang, KV caching, continuous batching, chunked prefill, quantization (AWQ/GPTQ), speculative decoding

Systems: Python, C++, FastAPI, AsyncIO, Redis, PostgreSQL, Docker, Linux, AWS, GCP, OpenTelemetry

Post-training: LoRA/QLoRA, TRL, DAPO/GRPO, process reward models, evaluation pipelines

EDUCATION & RECOGNITION

Dr. APJ Abdul Kalam Technical University

B.Tech, Computer Science | 2026

International Youth Math Challenge Gold Honour | Top 1% TypeScript Engineer Globally (Algora) | Amazon ML Summer School 2025